21072 DATA ANALYSIS MODULE I (DATA LAB FOR ENTREPRENEURSHIP)

Classe impartita in lingua inglese

Programs: INTENT (8 crediti - I sem. - OB | 8 cfu - ECON-04/A)
Class: 27 (I sem.)
Department: Management and Technology

Course director: STEFANO BRESCHI

Classe 27

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Attenzione! Il programma di seguito riportato è da considerarsi valido SOLO per l'anno accademico 2026/2027.

MISSION

This course aims to provide students with a strong foundation in the principles and techniques of data analysis and econometric modeling. The first part will cover topics such as data manipulation, data visualization, and statistical analysis using Python, one of the most popular programming languages. In this part of the course, students will learn how to write efficient and effective code to analyze and interpret data and will develop the ability to communicate their findings in a clear and concise manner. The second part of the course will cover statistical inference and regression analysis. Students will learn how to apply these techniques to real-world data sets and will develop the skills necessary to interpret and communicate their findings effectively. Throughout the course, a special emphasis will be put on the importance of critical thinking and problem-solving in the field of data analysis and econometrics. By the end of the course, students will be equipped with the knowledge and skills necessary to conduct rigorous

data analysis and econometric modeling in a variety of contexts in their future academic or professional pursuits.

COURSE CONTENT DETAILED - SCHEDULE

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In today’s data-driven landscape, understanding the dynamics of high-tech ventures requires mastery of complex, multi-source datasets and advanced analytical tools. This course empowers students to streamline and reshape raw startup ecosystem data—from fundraising rounds to performance metrics—into robust, analysis-ready formats. Through hands-on labs in Python and Stata, students will develop reproducible pipelines for data cleaning, integration, and visualization, then apply core econometric techniques (OLS, IV, panel models) to test hypotheses on growth, valuation, and investment trends. Throughout the lab, we emphasize critical thinking and problem-solving as essential skills for navigating complex data workflows and econometric challenges. By the end of the course, students will be fully prepared to carry out robust data transformations and econometric analyses across a range of academic and professional settings.

Class group 27 — INTENT — Breschi S. / Gagliardi L. — 10:15-12:45, Room N39, Sraffa 13 (3rd floor)

#	Day	Date	Time	Room	Topic
1	Monday	07-Sep	10:15 - 12:45	N39	Intro + Python recap
2	Friday	11-Sep	10:15 - 12:45	N39	Pandas DataFrames
3	Monday	14-Sep	10:15 - 12:45	N39	Pandas: subset, add, remove data
4	Monday	21-Sep	10:15 - 12:45	N39	Pandas: cleaning data, missing
5	Friday	25-Sep	10:15 - 11:45	N39	Pandas: grouping data
6	Monday	28-Sep	10:15 - 12:45	N39	Pandas: merging data
7	Friday	02-Oct	10:15 - 12:45	N39	Pandas: working with text
8	Monday	05-Oct	10:15 - 12:45	N39	Nature of econometric data

9	Friday	09-Oct	10:15 - 12:45	N39	Descriptive Analysis
10	Monday	12-Oct	10:15 - 12:45	N39	Linear regression model (1)
11	Friday	16-Oct	10:15 - 12:45	N39	Linear regression model (2)
12	Friday	30-Oct	10:15 - 12:45	N39	Hypothesis testing
13	Monday	02-Nov	10:15 - 12:45	N39	Further specification issues
14	Friday	06-Nov	10:15 - 12:45	N39	Binary dependent variables
15	Monday	09-Nov	10:15 - 12:45	N39	Data lab* / Guest speaker**
16	Friday	13-Nov	10:15 - 12:45	N39	Data lab* / Guest speaker**
17	Monday	16-Nov	10:15 - 12:45	N39	Data lab* / Guest speaker**
18	Monday	23-Nov	10:15 - 12:45	N39	Data lab* / Guest speaker**
19	Friday	27-Nov	10:15 - 12:45	N39	Final presentations
20	Monday	30-Nov	10:15 - 12:45	N39	Final presentations
21	Friday	04-Dec	10:15 - 12:45	N39	Final presentations

Class group 28 — INTENT — Breschi S. / Gagliardi L. — 13:45-16:15, Sraffa 13 (3rd floor)

#	Day	Date	Time	Room	Topic
1	Monday	07-Sep	13:45 - 16:15	N39	Intro + Python recap
2	Friday	11-Sep	13:45 - 16:15	N37	Pandas DataFrames
3	Monday	14-Sep	13:45 - 16:15	N39	Pandas: subset, add, remove data
4	Friday	18-Sep	13:45 - 16:15	N37	Pandas:

					cleaning data, missing
5	Monday	21-Sep	13:45 - 16:15	N39	Pandas: grouping data
6	Monday	28-Sep	13:45 - 16:15	N39	Pandas: merging data
7	Friday	02-Oct	13:45 - 16:15	N37	Pandas: working with text
8	Monday	05-Oct	13:45 - 16:15	N39	Nature of econometric data
9	Friday	09-Oct	13:45 - 16:15	N37	Descriptive Analysis
10	Monday	12-Oct	13:45 - 16:15	N39	Linear regression model (1)
11	Friday	16-Oct	13:45 - 16:15	N37	Linear regression model (2)
12	Friday	30-Oct	13:45 - 16:15	N37	Hypothesis testing
13	Monday	02-Nov	13:45 - 16:15	N39	Further specification issues
14	Friday	06-Nov	13:45 - 16:15	N37	Binary dependent variables
15	Monday	09-Nov	13:45 - 16:15	N39	Data lab* / Guest speaker**
16	Friday	13-Nov	13:45 - 16:15	N37	Data lab* / Guest speaker**
17	Monday	16-Nov	13:45 - 16:15	N12	Data lab* / Guest speaker**
18	Monday	23-Nov	13:45 - 16:15	N39	Data lab* / Guest speaker**
19	Friday	27-Nov	13:45 - 16:15	N37	Final presentations
20	Monday	30-Nov	13:45 - 16:15	N39	Final presentations
21	Friday	04-Dec	13:45 - 16:15	N37	Final presentations

TEACHING METHODS

Selection:

- Lectures
    - Lectures
- Guest speaker's talks (in class or in distance)
- Practical Exercises
    - Software-Based Exercises
    - In-class exercises
    - Data Analysis, Graphic Representations/Calculators
- Individual works / Assignments
    - Project Work
    - Project Presentation
- Collaborative Works / Assignments
    - Report/essay
    - Project work
    - Project Presentation

The course adopts a hands-on, bootcamp-style approach that combines lectures, coding sessions, and applied econometric analysis. Almost every session integrates the explanation of concepts with their direct implementation on real-world data.

- **Lectures and conceptual Inputs**: Introduction to Python for data wrangling and visualization, and to econometric methods (using Stata) for regression analysis and hypothesis testing.
- **Practical, software-based exercises**: In-class coding labs where students learn to import, clean, structure, and analyze data using Python and econometric software.
- **Guest speaker seminars**: Industry practitioners present applications of data to startup finance, venture capital, and market analysis.
- **Collaborative project work**: Students work in teams to develop an empirical project based on Pitchbook data (US startups, 2012–2019). Teams identify a research question, construct a dataset, apply econometric methods, and present results at the end of the course.
- **Team support and feedback**: Dedicated sessions with the teaching assistant to guide teams on dataset construction, model selection, and interpretation of results.
- **Problem-solving and self-assessment**: Instructor-led exercises, mock exams, and quizzes to reinforce core techniques and ensure continuous learning.

This blended approach ensures that students acquire both the technical skills (Python programming, econometric modeling) and the applied research experience needed to turn complex data into actionable business insights.

ASSESSMENT METHODS

Selected methods:	Continuous assessment	Partial exams	General exam
Written individual exam (traditional/online)			**x**
Open and/or closed answers questions exam			x
Collaborative Works / Assignment (report, exercise, presentation, project work etc.)	**x**		
Assessment of the results achieved/outcome	x		

- **Attending students**
    - *Written Exam (60% of final grade)*: Duration 1.5 hour. Includes both open and closed questions, as well as multiple-choice items. The exam evaluates students' understanding of econometric methods, ability to interpret results, and capacity to apply them to business problems.
    - *Group Project (40% of final grade)*: Teams of up to 5 students develop an empirical project using Pitchbook data (U.S. startups, 2012–2019). Projects must identify a research question, build and analyze the relevant dataset, and present findings in a slide deck (about 25 minutes). All group members receive the same grade. To be considered "attending," students must participate in a group and deliver a final presentation. The project grade remains valid for all exam sessions in the academic year.

- **Non-attending students**
    - *Written Exam only (100% of final grade)*: Duration 2.5 hours. Includes open, closed, and multiple-choice questions. Focuses on econometric techniques and

their application to real-world data, with no project component.

- **Organizational details**
    - Group self-enrollment is managed via Blackboard.
    - Project support sessions are organized with the teaching assistant to guide students on dataset construction, econometric methods, and interpretation of results.
    - Mock exams will be distributed during the course to familiarize students with exam structure.
    - The written exam is administered on Blackboard with Respondus Lockdown Browser.

- **Assessment criteria**
    - Written exam: accuracy of answers, clarity of reasoning, and ability to link methods to applications.
    - Project: originality and clarity of the research question, appropriateness of empirical methods, quality of data analysis, and effectiveness of the presentation.

The final grade is determined by the weighted average of the written exam and project for attending students, and by the written exam only for non-attending students.

GENAI USE

The use of Generative Artificial Intelligence tools by students, solely as support during teaching activities*, is:

- Consentito
- Allowed

TEACHING MATERIALS

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Attending and non-attending students

- *Part I (Python and Data Analysis)*: Jupyter notebooks distributed in class.
    - (Optional textbook) Stephanie Molin (2021). *Hands-On Data Analysis with Pandas*. Packt Publishing Limited (chapters, 1-4).
- *Part II (Econometrics)*: Lecture slides and a curated selection of research papers made available in advance through Blackboard.
    - Hill, R. C., Griffiths, W. E., & Lim, G. C. (2018). *Principles of Econometrics*. John Wiley & Sons (selected chapters).

Sharing and access

- All teaching materials (slides, datasets, Jupyter notebooks, project instructions, and readings) will be distributed via Blackboard.
- Auxiliary files for project work (data dictionary, ER diagram, etc.) are available in the dedicated Pitchbook folder on Blackboard.
- Selected bibliographic references are accessible through the Bocconi Library (Course Reserve).

HONOR CODE

Bocconi University conceives of education as an ongoing process that stretches across a person's entire professional life. The University hopes that the entire Bocconi community will respect the values of fairness and correctness associated with it, values which inspire and guide the conduct of all community members as they pursue common objectives and a shared mission. Bocconi University is also committed to building a truly inclusive climate based on diversity. With the collaboration of faculty and students, it aims to ensure a safe learning environment free from any form of discrimination based on race, religion, gender, geography, age, disability or sexual orientation, in which there is room for a diversity of approaches and where discussions are always conducted in a respectful manner. The Bocconi University Honor

Code is published on the website at http://www.unibocconi.eu/honorcode, we encourage all students to read it.

COURSE ETIQUETTE

AI Policy

In this course you are strongly encouraged to use AI, but always with a clear purpose. Think of AI as an extra team member you can question and challenge: use it to test your ideas, explore different perspectives, and refine your reasoning. You may find it helpful both when preparing your project and when studying the course material more broadly. In both situations, the goal is not to let AI provide ready-made answers, but to use it as a tool to deepen your understanding and strengthen your ability to explain and defend your own reasoning. This will also prepare you for the final exam, which you will complete without access to AI. Whenever you use AI in preparing your work, you must include a short note in your document to acknowledge how it was used.

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